Tasnem Moura | tasnmmoura@gmail.com | Linkedin | tasnemmoura.vercel.app

Summary:

Electrical and Computer Engineering graduate with a strong academic background in VLSI systems, microelectronics, digital design, and hardware development. Hands-on experience in complex engineering projects involving circuit design, simulation, validation, and system analysis. Strong analytical and problem-solving skills with a passion for semiconductor technologies and multidisciplinary hardware engineering environments.

Education:

- **2020–2024: *The Hebrew University of Jerusalem , B.Sc. in Electrical and Computer Engineering***
  - Focus: Microelectronics, VLSI Design, and Digital System
  - Core Courses: The Physics of Semiconductor Devices, Advanced VLSI, Interfacing Analog and Digital Worlds (Verilog), A nalog Electronics Lab, Python, Data Structures, Micro-Electronics Devices Lab, Engineering Project Workshop.

Certificates :

- ***2026 : AI Engineer Career Accelerator Program***
  Led by Mohammad Kabajah, Sr. AI/Cloud Infrastructure Architect at AWS
- **2025: *NVIDIA Deep Learning Institute (DLI)*** – Accelerated Data Science Workflows; Introduction to CUDA
- **2024: *Tel Aviv University*** – Business Economics, People & Systems Management, Statistical Models
- **2023: *Bridgz Company* -** Three-month theoretical and practical training programs focused on chip design and RTL-to-GDSII flow :
  - Industry -Oriented Training in VLSI (Physical Design)
  - Chip Design Verification
- **2022: *Technion*** *Israel Institute of Technology*
  - All-Digital Phase-Locked Loops Course

Projects:

- ***AsyncProof – AI-Assisted Formal Verification of Asynchronous Controllers (in progress)*** –
  - Building an open-source tool that converts timing diagrams of asynchronous controllers into **Signal Transition Graphs** and verifies them formally with **Workcraft** (Python, Docker)
  - Built an independent STG checker cross-validated against Workcraft; **247 automated tests**
  - Showed that one misread causal arc passes all formal checks yet yields a different circuit
- ***Two-Step Time-Domain ADC with Delay-Tracking SAR TDC*** –
  - Designed and simulated a **6-bit Time-Domain ADC** in **14nm CMOS** (Cadence Virtuoso)
  - Implemented Sample & Hold, VTC, and Delay-Tracking SAR TDC blocks
  - Performed transient and corner simulations, investigated timing behavior, and validated performance against design s pecifications.
  - Achieved **1.25 GS/s, 7.4 mW, 50 dB SNDR**
- ***MEMS-Based Optical System*** –
  - Designed and analyzed an electromagnetically actuated **MEMS micromirror** for high-speed beam steering.
  - Developed electro-mechanical models using COMSOL and MATLAB and validated simulation results against analytical calculations.
  - Achieved ±10° tilt, >100 kHz resonance, and <10 µs response time
- ***AI-Powered Recruitment Platform (AI Engineer Career Accelerator, in progress)*** –
  - Developing the platform for CV analysis, candidate-job matching, gap identification, and tailored recommendations.
  - Applying **LLMs, agentic AI concepts, document processing, information retrieval, and AI-assisted development** within a team-based engineering project.

Experience:

- **2020–2023:** *Tutor*, The Hebrew University of Jerusalem – Classical Mechanics, Analog & Digital Electronics
- **2020–2022:** *Shift Manager*, Bloomfield Science Museum – Delivered interactive science workshops, explained scientific concepts through a wide range of museum exhibits, trained museum guides, and supervised daily shift operations.

Technical Skills:

Digital Design & Verification: Verilog / SystemVerilog, RTL simulation, debugging & root-cause analysis , Functional valida tion, test, scenario development, waveform analysis ,Understanding of verification flows, correctness checking, and ASIC des ign concepts

***VLSI & Systems*:** RTL-to-GDSII flow understanding, digital/analog circuit fundamentals ,SoC architecture concepts, power-e fficient design considerations ,Microelectronics & semiconductor device fundamentals

Programming & Scripting: Python, C++, MATLAB, C, C#, SystemVerilog

Hardware & Lab Tools: Cadence Virtuoso , COMSOL Multiphysics , Oscilloscopes, logic analyzers, spectrum analyzers, sign al generators

Industry Conferences:

- **2021–2026: *50+ industry conferences* –** including WO.men CEO Summit (2025, 2026), FV-Israel (Intel), START @Google – Physical Design Bootcamp, Annapurna Labs (AWS), From Models to Hardware (8200 Alumni), Code Name: Woman (Micr osoft), AWS Experience and NVIDIA.

Community Involvement:

- **2022: *Educational Assistant*, *Shams Association* –** *Facilitated cognitive intelligence games for children, observed and do cumented their behaviour, and supported personalized learning strategies.*
- **2021–2022: *Chess Instructor*, Chess Association –** Taught structured chess lessons to enhance students' strategic thinkin g, focus, and problem solving skills.

Languages:

| Hebrew – Fluent | English – Fluent | Arabic – Native|